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| | Our hypothesis is that collagen molecules are capable of different ‘modes’ of material (liquid crystalline) self-assembly, some of which order collagen solutions prior to triple helix formation. This amphiphilicity and pre-collagen lyotropic phases may be as important in structure formation as liquid crystals formed by rigid triple helices. An understanding of the interplay between lyotropic soap-like behavior, rigid rod packing behavior and thermotropic types of liquid crystallinity and triple helix melting folding is expected to have profound implications for collagen assembly in vitro and in vivo. Our approach is twofold, (1) compare a series of peptides with identical overall chemistry and different triple helix stabilities - stable, unstable, sterically precluded triple helix - to determine which phase features are due to soap-like behavior and which require the triple helix, (2) use different endblocks attached to the same stable triple helical sequence to systematically change the amphiphilic interactions (hydrophilic charged interactions, hydrophilic polar interactions, non-amphiphiles or weakly amphiphilic molecules). This allows us to compare rigid rod orientated | | |

liquid crystalline phases with amphiphilic interactions that are repulsive, attractive, absent and determine what aspects of structure/phase behavior REQUIRE amphiphilicity. Investigation of phase behavior/morphology development in regions where triple helix is folding will allow investigation of coupling of two types of interactions driving phase behavior – a simple model for biocomplexity. The value of the research will be in understanding development during triple helix folding and this insight is expected to depend sensitively on a number of parameters. This insight will be a good model for control of biocomplexity, understanding/insight into general chemical features leading to complex biological structures and patterns, tools for biomimetic materials design, effects such as gravity driven sedimentation may be dampened or amplified depending on other environmental parameters, understanding of bone loss problems in space as collagen serves as the primary matrix for hydroxyapatite nucleation and growth.
There have been numerous studies on the spontaneous ordering of biological molecules. The overwhelming majority of the research has focused on the binding activity of specific sites – molecular recognition. While specific recognition between biological molecules is inarguably a very important aspect of biomolecular structure formation, it is not the only piece in the puzzle. In order for site-specific recognition and binding to occur, sites have to first be in proximity. An organizing process prior to solidification, recognition, and binding is implied; the underlying phenomenon would have to reliably and robustly organize biological molecules into ordered domains. Liquid crystallinity immediately suggests itself as a likely mechanism for the portion of the self-assembly process that occurs prior to specific recognition and binding. In fact, liquid crystallinity and liquid crystal–like textures have been observed for biological molecules, especially fibrous proteins such as collagens. An in depth understanding of the forces controlling collagen hierarchical assembly will provide new avenues for the formation of collagen-based biomaterials for a variety of NASA-related needs. Additionally we expect that microgravity will provide advantages to the study of biomacromolecular hierarchical self-assembly. In a microgravity environment, effects strongly influencing the liquid crystalline morphology and that of the materials fabricated from liquid crystals, such as surface anchoring and density fluctuations, can be eliminated. Because the precursor phases used to make liquid crystalline morphologies and materials are fragile, even weak forces such as gravity can have a strong influence on domain sizes and organizational morphology. Furthermore, unlike crystal growth, order in these fragile precursor phases are formed and stabilized cooperatively, rather than through nucleation and growth of a highly stable ‘hard’ domain as in crystallization. This organization within a domain can be influenced within ‘nuclei’ or domains of the ordered phase, as well as during growth. This combination of factors suggests that the direct influence of gravity may have noticeable effects on the domain structure morphology and even the phase of the biological liquid crystalline materials generated. Indirect effects are expected to include container wall interactions as grains or domains of the ordered phase settle due to density differences.

We expect to obtain large well-organized domains of fibrous collagen. Ideal liquid crystalline grains formed in this manner would be invaluable in understanding the factors driving and controlling self-assembly. Various types of fragile liquid crystalline behavior have been observed in our prior studies; the formation of compex fluid structures such as micelles, due to amphiphilic interactions, and orientation of rigid rod molecules into chiral condensed concentrated phase liquid and plastic crystals. Our evidence suggests that flexible amphilphilic behavior of unfolded collagen sequence peptides can organize the oligopeptide monomers prior to folding and aggregation into triple helices (and rigid rods), at lower concentrations and higher temperatures. A number of highly chiral morphologies are also observed when the molecules are folded into triple helices. We believe that these two behaviors or phase regimes overlap and that the presence of both organizing tendencies may enable the growth of very large homogeneous domains. The coupling of two or more self-assembly processes may also explain the facility with which these model molecules both form large liquid crystalline domains (a process impeded by high concentration/low mobility) and solidify while preserving the liquid crystalline domain texture (a process aided by high concentration/low mobility). This contrasts with small molecules, which have a high mobility and rearrange to a new phase (or collapse) unless solidified very rapidly.

Our objective is to systematically investigate the influences of micellar order in unfolded triple helices and of orientated liquid crystalline order in triple helices by decoupling the two behaviors through specific sequence designs. The major plans are as follows:

(1) Synthesis and Characterization of Model Collagens - To gain further insight into assembly and patterning of collagens, we will synthesize an additional series of collagen analogs in which we will explore: (a) the influence of endblock hydrophobicity/hydrophilicity on assembly. Peptides (~30 residues) will be prepared by peptide synthesis and larger proteins (10-15 kDa) for multi-domain assemblies will be synthesized as recombinant proteins generated in E. coli and yeast (P. pastoris).

(2) Characterization/Fingerprinting of Collagen Conformations – The model collagens (peptides and recombinant proteins) will be characterized for conformational behavior and transitions in order to correlate molecular sequence and environment with geometry (shape in 3D) and conformation behavior related to chiral morphology features.

(3) Characterization/Fingerprinting of Phase Behavior – The phase behavior of the model peptides and recombinant proteins will be characterized as a function of: (a) primary sequence of peptide/protein, (b) concentration, and (c) temperature. Flexible lyotropic (unfolded oligopeptides) phases and rigid rod (collagen triple helix) phases and morphologies will be studied.

(4) Details of Textured Materials - The effect of lyotropic and rod-like liquid crystalline order overlap/coupling on details of liquid crystalline textures/solid materials generated from these phases will be studied. Detailed physical characterization of materials will include: (a) domain size, (b) orientation, and (c) defect structures.

(5) Engineered Materials for Applications - Environmental parameters will be manipulated to engineer materials for specific applications related to biomaterials based on collagens.

**Task Description:**

Collagen is the most commonly used natural biomaterial, whether reprocessed from bovine sources, generated as genetically engineered variants, or reconfigured as gelatin-based matrices. Collagen is structurally complex due to hierarchy at different length-scales and this structural complexity forms the basis for the diverse mechanical and biological features of collagen that vary with each tissue type. There is an exhaustive literature on fibrillar collagens, including several characterized by detailed structural models. However, there is little experimental data available describing the role and mechanism of the helical segments formed from collagen proteins in controlling the size and shape of the assembled fibrils. Insight into these issues is critical in order to design biomaterials in vitro based on collagen to match mechanical and biological profiles in vitro and in vivo, and to understand collagen processing and assembly in vivo. Liquid crystallinity is the likely mechanism for the portion of the self-assembly process that occurs prior to specific recognition and binding related to fibril organization. In fact, liquid crystallinity and liquid crystal–like textures have been observed for fibrous proteins such as collagen; reported as helicoids. An in depth understanding of

the forces controlling collagen hierarchical assembly will provide new avenues for the formation of collagen-based biomaterials for a variety of NASA-related needs. Additionally, microgravity will provide advantages to the study of biomacromolecular hierarchical self-assembly in collagen-based biomaterials since the effects strongly influencing the liquid crystalline morphology and that of the materials fabricated from liquid crystals, such as surface anchoring and density fluctuations, can be eliminated.

Our objective is to systematically investigate the influences of micellar order in unfolded triple helices and of orientated liquid crystalline order in triple helices by decoupling the two behaviors through specific sequence designs. We will extend our observations from ongoing studies into the hierarchical assembly process to gain a stronger foundation of the fundamental control points in assembly paths related to concentration, temperature and specific phases. We will continue to use our mini-triblock system for the collagen-like peptides and proteins to facilitate the studies as this approach has provided a solid basis for gaining qualitative and quantitative insight into relationships between primary sequence and structural hierarchy. We will continue to assess the structures with a variety of analytical tools including polarizing optical microscopy, atomic force microscopy, field emission scanning electron microscopy, fourier transform infrared spectroscopy, circular dichroism, x-ray diffraction, and synchrotron x-ray scattering. The major plans for the study include: (1) synthesis and characterization of model collagens to explore the influence of endblock hydrophobicity/hydrophilicity on the assembly - both synthetic peptides and genetically engineered longer versions of the peptides will be utilized, (2) characterization/fingerprinting of collagen conformations and transitions in order to correlate molecular sequence and environment with geometry (shape in 3D) and conformation behavior related to chiral morphology features, (3) characterization/fingerprinting phase behavior, (4) details of textured materials including domain size, orientation, and defect structures, and (5) formation of engineered materials for specific applications related to biomaterials based on collagens. Various avenues exist to manipulate the hierarchical self-assembly process of collagen, both in bulk and interfacial materials and their precursor phases. The different control points useful in manipulating self-assembly depend on end-group and core sequence chemistry and interactions, which are the subject of our scientific studies. Thus there is a close connection between developing an understanding of the bases of biomimetic self-assembly and in engineering parameters to control nanostructured soft materials, particularly based on collagen-based biomaterials. The outcomes of the proposed studies will include: further refinements in the processing paths related to mesophases, in particular, the quantitative aspects of concentration-dependence and temperature-dependence on the transitions and phases formed; understanding of the influence of molecular weight on the phase diagrams (e.g., comparisons between the shorter peptides and the longer recombinant variants of the collagen sequences); an ability to predict structural features and patterning of collagen-based biomaterials from sequence and environmental factors; a broader understanding of the role of molecular recognition in combination with environmental factors on structural hierarchy; and new insight into collagen-based biomaterials applied directly to a variety of needs. A comprehensive training and education plan is included for the students involved in the program that is founded upon interdisciplinary themes related to biology and materials science.

Studies into collagen-based materials have far reaching implications for a wide range of materials, biomaterials and biological interactions. Collagens are found in most tissues in the body. In addition, they are routinely used in reprocessed forms to generate biomaterials and scaffolds for cell and tissue growth. Despite this widespread use, there has been little effort to understand the details of the self-assembly process in vitro. This insight will be critical to the full understanding of biological responses to collagen-based materials, potential to fabricate collagen-based biomaterials. Collagenous materials are produced commercially for a variety of biomedical uses. These products, including collagen sponges and fibers, are used for autologous tissue regeneration, suture materials, cell growth matrices and related needs. Currently, the structural features of these materials are controlled at the macroscopic level. For example, the pore sizes in collagenous membranes and sponges are usually determined by removal of water in the form of ice from frozen samples. There is little attention paid to lower levels of structural organization in these materials, including the molecular and mesoscale levels. These levels of organization are critical to the mechanical properties and biological functions of these materials. We also believe that in order to fully utilize collagenous materials for biomaterial and tissue engineered products, a better understanding of the hierarchical assembly of these materials in vitro will be essential. For example, while calcium loss from bones during exposure to microgravity is a major problem, it is perhaps reasonable to consider that the assembly or remodeling of collagen in bone as the template upon which calcium phosphate nucleates and grows as a potential contributor to this problem.

In the broader context, collagenous matrices play a large role in cell growth, cell regrowth, and cellular differentiation. The question of how collagen molecules form collagenous tissues is of fundamental importance to the biological sciences. Extracellular matrices form templates for subsequent cell growth, incursion and differentiation. Cell growth and morphology are affected by the patterning of the cell growth substrate, and collagenous extracellular matrix morphology or patterning may influence cell differentiation. For some non-biological materials used as biomaterials, regular surface features or “microroughness” with a periodic arrangement and length-scales in the 1 ?m to 5 ?m range appear to have a pronounced effect on cell spreading and morphology. It is interesting that these relevant periodicities are bracketed by the cholesteric periods observed for the collagen model peptides and other fibrous proteins.

In summary, aside from direct impact on collagen-based biomaterials processing and utility, NASA will benefit from the proposed studies in a number of ways:

Physics – The raw data can be easily obtained for insight into multiple coupled phase orientations and frustration due to competing ordering tendencies

Materials Science – Information on how to make nanopatterned materials from chiral liquid crystals – large domains and easy solidification., including routes to biomimetic processing of films and fibers.

Biophysics – Interaction of key self-assembly processes generate many ordered structures identical to structures in vivo. How do coupled/competing phase transitions drive biological self-assembly (e.g. protein folding in general)?

Biomedical – Extracellular matrix patterned materials with controlled orientation on relevant length scales for cell responses.

Microgravity – Fabrication of materials in space (biosynthetically), with medical relevance for organization of collagenous tissues in zero gravity that may be pertinent to bone mass loss.

**Rationale for HRP Directed Research:**

**Research Impact/Earth Benefits:**

**Task Progress:**

Task Progress - Year 3
Collagen Peptides for Assembly and Mineralization Patterns – Relationships Between Primary Sequence, Assembly and Function - The collagen mimetic peptides used in the studies of collagen assembly were described in year 2 and were modified to include terminal cysteine residues for surface anchoring. In order to improve the solubility, some polar amino acids including arginine, lysine, glutamic acid and aspartic acid were introduced into the sequence. The peptides were capped with cysteine at the N-terminus to facilitate adhesion on gold surfaces. Gold nanoparticles were prepared with the reduction of chlorauric acid by trisodium titrate. Mineralization was carried out by soaking collagen-like peptides films in 200 mM aqueous calcium chloride solution with above tris buffer (Ca solution) for 2 hrs at 37 ºC. The films were studied by atomic force microscopy (AFM), charged coupled device (CCD) camera and X-ray photoelectron spectroscopy (XPS).

The citrate crystals grown on the native collagen-like peptide coated gold nano particle were imaged. The platelet-shaped crystal chips with irregular edges were observed in AFM phase image. Compared with citrate crystals on native collagen peptide, the size was increased over 100 nm on OI-A coated particles. The possible reason could be the distorted triple helices due to the alanine substitution which resulted in the loose mineral deposition. The structural analyses of the resulting crystals are needed in future studies. The OI-V and OI-D peptides were also investigated by similar procedure. No crystals were observed on particle surfaces. The gold particles aggregated and spread on the substrate surface to form a relatively smooth layer. Only large crystals with larger than 50 μm in diameter were observed. This indicated the OI-V and OI-D were not suitable matrices for citrate deposition. The citrate was crystallized not on peptide films but in solution between particles. The profile was similar to that of citrate crystals without the presence of collagen-like peptides.

The citrate stabilized gold nano particle aqueous solution was mixed with 1 mg/ml collagen-like peptide solution at volumetric ratio of 2:1 in the similar procedure as described above. In order to observe large citrate crystals, the mixture was dropped on surfaces and placed in a semi-closed system allowing slow dry (usually overnight) instead of air or blow dry. The obtained crystals were imaged by CCD. Thicker and larger crystals were observed in the case of native collagen-like peptide compared to other OI peptides. In the case of OI-D, only small crystals were observed which were not comparable in size and density with respect to other OI peptides. The observed crystal size was in the sequence of native > OI-A > OI-V > OI-D, which was consistent with the severity of the OI with different glycine substitution. Similar trend was observed on gold surface. Due to the different affinity between the particle/crystal and the substrate surfaces, the citrate crystals exhibited a ramiform profile on gold surfaces. The size of the crystal and the branch density decreased with respect to the above trend. The citrate stabilized gold nano particles without the presence of collagen-like peptides were also treated in the same manner on silicon and gold surfaces as control experiments. Globular crystals were formed on silicon surface and leaf-shaped crystals were observed on gold surface, indicating those above observed crystals were formed with the interference of collagen-like peptides and the exhibited crystal profiles were related to the natures of the peptides.

Besides self-assembled on gold nano particles, the collagen-like peptides were also studied on direct assembly on gold surfaces. A DI Dimension 3100 scanning probe microscopy (SPM) with Nanoscope IV controller, which equipped with a RTESP probe (spring constant of 300 kHz), was used to generate high aspect ratio images. From the topography images, the native and OI-A collagen peptides exhibited relatively ordered surfaces with smaller globular peptides. With the increased size of the residue groups, the OI-V and OI-D exhibited a loose deposition on gold surface with larger peptide clusters. This was possibly due to the partially untied triple helices and severe distortion of the molecules which resulted in the improper assembly on the surface. Since the peptides were capped with cysteine, the molecules were affixed to the gold surface at one end. The assembly was strongly affected by the shape of the molecules. Distorted chain would be less likely to associate with the other or only a partially loose assembly to form bulky clusters. Based on the phase images, the uniformity of the peptide films also exhibited a similar trend as observed above.

The thio-Au bonding would have some effects toward the peptide assembly on surface. In order to observe their free assembly behavior, the silicon surfaces were used to physically absorb collagen-like peptides. The silicon chips were immersed into 1 mg/ml peptide solution for 30 minutes followed by thoroughly rinsing with ultra pure water to remove any unbonded molecules. The adsorbed peptide layers were imaged by AFM in typing mold. Due to the lack of covalent bonding to the surfaces, the collagen-like peptides were adsorbed on silicon by hydrophobic interactions and allowed to free assembly. The native collagen peptide adsorbed surface exhibited an ordered artifact. OI-A was not as remarkable as native one. Almost no assembly was detected on OI-V and OI-D adsorbed surfaces. This observation demonstrated that the assembly ability was in the order of native > OI-A > OI-V, OI-D and was consistent with the severity order of OI.

The collagen-like peptide monolayers were prepared by immersing gold surfaces in 1 mg/ml peptide aqueous solution for 10 minutes followed by thoroughly rinsing with ultra pure water to remove unbonded peptides. The collagen-like peptide SAMs were used as matrices for the hydroxyapatite mineralization. AFM phase images were recorded for the first, second and third rounds. With the progress of the mineralization, nucleation occurred and minerals grew on all types of peptide films. The growth rate became faster after three rounds. The size of the minerals exceeded the range of AFM scanner and could be seen by eye. The changes in the mineral size and density were significant on native collagen peptide SAMs. The minerals on native one were larger and thicker than those of on OI peptide SAMs which attributed to the ordered assembly facilitating the mineral growth. It was difficult for minerals to nucleate and grow on severe mutated collagen-like peptide films like OI-V and OI-D. Unlike native one on which platelet crystals formed, OI-A would result some large clusters of minerals consisted of many small crystalline. The composition of the obtained minerals needs to be further characterization. XPS will be used in future studies to obtain the atomic ratio of Ca/P which could determine if the crystalline hydroxyapatite or amorphous calcium phosphate formed during the mineralization.

In related but separate studies, cell adhesion plays a central role in a wide variety of biological processes. Cells adhere to extracellular matrix in order to maintain the architecture of specific tissues and organs. Proliferation, differentiation, aggregation or even apoptosis depend on the interactions of cell-adhesive receptors and different substrates that facilitate the engagement of a cell to the extracellular matrix or to other cells. The materials that can not only attract and enhance the growth of the particular cells but also facilitate the ingrowth of preferred cells are in high demand. All cells have important cell-matrix adhesion receptors that facilitate the communication between cell and matrix. However, there

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| | is little understanding of how the changes in collagen structure have influence on the role of the receptors' primary motifs involved in the direct linkage with extracellular matrix. Additionally, there is no information when silk is used as the matrix for such interactions. The primary goal was to identify sequences that specifically interact with biomatrices such as collagen, and in particular how the structural features of collagen impact these interactions. To investigate the cell-matrix adhesive motifs, the random peptide phage library was used. The phage library mimics the shape and chemical composition of peptides that come in close contact with matrix proteins. This provides an opportunity to identify the critical contact residues that contribute to the cell-matrix interactions. The set of experiments, called biopanning, was conducted on: native collagen, denatured collagen and tissue-culture plastic. The results indicate that both native and denatured forms of collagen type I attract various structural motifs. These motifs are different to those attracted by tissue-culture plastic. Such sequences can mediate the adhesion of cells to the extracellular matrix or engineered matrices. They also can to propagate various cell signals. Further investigation is ongoing to assess quantitative aspects of binding affinity, specificity and biological relevance of these sequences. |
| **Bibliography Type:** | Description: (Last Updated: 02/01/2019) |
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